Objective:

To provide clinical guidance on management of children who are contact of contagious Tuberculosis (TB) patient

Policy Statement:

- Management of children with TB Contact

Policy Scope:

Health care workers including Doctors and nurses taking care of pediatric patients who are contact of contagious TB patients

Introduction:

- TB is an infectious disease usually caused by a bacterium called MTB. The bacteria usually attack the lungs, but TB bacteria can attack any part of the body such as the kidney, spine and brain. Not everyone infected with TB bacteria becomes sick. As a result, two TB-related conditions exist: LTBI and TB disease. If not treated properly, TB disease can be fatal.

- Children under the age of 5 years are at high risk of progression to severe forms of TB disease after TB infection.

- The manifestations of TB infection, after exposure to a person with active infectious TB, are very different in infants and children compared to adults due to following reasons:
  - First, the diagnosis of TB disease in children under 5 years of age can be difficult because they often have nonspecific signs and symptoms and a paucity of *mycobacteria*.
  - Second, TB in children is considered a sentinel event usually indicating recent transmission.
  - Third: children, especially infants, are at an increased risk of progressing from latent TB infection (LTBI) to active and are prone to develop severe TB disease.
- Infection with MTB is usually the result of inhaling into the lungs infected droplets produced by someone who has laryngeal TB or open pulmonary TB and is coughing.
- The immune response (delayed hypersensitivity and cellular immunity) develops about 4–6 weeks after the primary infection. In most cases, the immune response stops the multiplication of TB bacilli at this stage. However, a few dormant bacilli may persist.

Risk of TB disease in children after contact with active TB:

- The risk of developing disease after infection is much greater for infants and young children under 5 years of age than it is for children aged 5 years or above (Table 1)
- This risk of developing TB disease after infection is greatest in the first two years.
- Isoniazid (INH) preventive therapy for young children with TB infection who have not yet developed disease will greatly reduce the likelihood of TB during childhood with a risk reduction of 70–90%.

Average age-specific risk for disease development after untreated primary infection

Age at primary infection	Manifestations of disease	Risk of disease (%)
<12 months	No disease	50
	Pulmonary disease	30–40
	TB meningitis or miliary disease	10–20
12–23 months	No disease	70–80
	Pulmonary disease	10–20
	TB meningitis or miliary disease	2–5
2–4 years	No disease	95
	Pulmonary disease	5
	TB meningitis or miliary disease	0.5
5–10 years	No disease	98
	Pulmonary disease	2
	TB meningitis or miliary disease	<0.5
>10 years	No disease	80–90
	Pulmonary disease	10–20
	TB meningitis or miliary disease	0.5

TB = tuberculosis.

Adapted from Marais et al.[7]

Terminology:

- Close contact: A child living in the same household as a source case (e.g. the child's caregiver) or who is in frequent contact with an index case
- Contagious or infectious TB: Pulmonary TB or laryngeal TB.
- Exposure: A situation in which a child has significant contact with an adult infected with

a contagious form of TB. A child exposed to MTB does not necessarily become infected. The recommended minimum period between the most recent exposure and tuberculin skin testing should be 10-12 weeks.

- Exposure history and time frame: a date three months prior to the date of diagnosis is to be assigned as the starting date for infection and this duration should be considered as the infectious period.
- Latent TB infection (LTBI): A condition in which MTB has entered the body and typically has elicited immune responses. LTBI is an asymptomatic condition that follows the initial infection; the infection is still present but is dormant and believed not to be currently progressive or invasive. The only sign of TB infection is a positive reaction to the TST or a positive result with the currently available blood test—the IGRA. For practical purposes, a child with LTBI is someone with a positive TST, no clinical evidence of disease, and a CXR that is either normal or demonstrates evidence of remote infection, such as a calcified parenchymal nodule and/or a calcified intrathoracic lymph node.
- TB disease: The diagnosis is based on symptoms, radiological changes and microscopy. Positive culture results for MTB are typically interpreted as both the indication and the confirmation of TB disease. For persons whose immune systems are weak, especially those with human immunodeficiency virus (HIV) infection, the risk of developing TB disease is significantly higher than it is for persons with normal immune systems.
- Multidrug resistant TB (MDRTB): This is defined as TB caused by MTB resistant in vitro to the effects of isoniazid and rifampicin, with or without resistance to any other drugs

Diagnosis

History:

- History of contact with contagious pulmonary TB, sputum smear sensitivity result of index case (if known).
- Children may be asymptomatic during initial period of infection
- Any symptoms or signs suggestive of TB: more than two weeks of low-grade fever with persistent cough unresponsive to antibiotics, loss of weight/failure to thrive (for the last three months) and lethargy.
- Other symptoms include haemoptysis, chest pain and shortness of breath.
- Local symptoms of affected specific site such as joint swelling, spine deformity and meningitis.

Physical examination:

General: growth assessment, fever, tachypnoea significant cervical lymphadenopathy.

Systemic examination for any signs of TB disease

GENERAL TESTING CONSIDERATIONS

Both the TST and IGRA depend on the host immune response to specific antigens found in M tuberculosis.

Sever TB disease can lead to immunosuppression and reduce the sensitivity of immune response-based tests such as the TST and IGRA. Consequently, both tests may have low sensitivity for tuberculosis.

<u>Tuberculin skin test (TST):</u>

- TST is the intradermal injection of purified protein derivative (PPD) which contains dozens of TB antigens
- Interpretation of the TST requires that the patient come for test reading after 48 to 72 hours.
- The measurement of the induration's <u>transverse diameter</u> should be made and documented to the closest millimetre.
- Interpretation of TST results in BCG recipients who are known contacts of a person with TB disease or at high risk for TB disease is the same as for people who have not received BCG vaccine
- TST (Mantoux test) is ≥5mm:it should be considered as positive in the following situations:

1. Children in close contact with known/suspected contagious cases of TB
2. Children suspected to have TB disease; findings in CXR consistent with active/previously active TB
3. Clinical evidence of TB disease
4. Children with HIV infection or immune deficiency states
5. children receiving immune suppressive therapy including immunosuppressive doses of corticosteroids (>20mg/day or >2mg/kg/day for two weeks or more) and anti-neoplastic agents.

*TST (Mantoux test) is ≥10 mm: it should be considered as positive in the following situations:

1- Children with frequent exposure to adults at high risk
2- Birth in or recent immigration (<5 years) from a high-prevalence country
3- Children who had travelled or been exposed to visitors from high-prevalence countries
4- Children with malnutrition, chronic renal failure, diabetes mellitus or lymphoma

*** IGRA: (interferon gamma releasing assay):**

- IGRAs are ex vivo blood tests that detect interferon-c (IFN-c) release from a patient’s CD4 and CD8 T lymphocytes after stimulation by antigens found on M tuberculosis complex (which includes M tuberculosis, M bovis, M africanum, M microti, and M canetti)
- IGRAs do not distinguish between TBI and TB disease

- Positive IGRA when done is equal to positive TST in guiding approach but negative test difficult to interpret and does not role out infection nor disease in children and immunocompromised patients

Chest x-ray:

- **Good quality anterior-posterior and lateral chest x-rays are required for the initial evaluation of TB disease in children**
- Child with TB disease may have non- specific findings in the chest -ray

Management:

The goal of latent TB treatment is to prevent TBI from progressing to TB disease and to diminish the reservoir for future TB cases

Management of Newborns of active TB disease mother:

- Newborn of a mother with smear positive pulmonary TB at delivery is at high risk of infection/ disease.
- If a pregnant woman has been on anti-TB (ATT) for at least 4 weeks before delivery and smear negative; it is low risk that her baby will become infected.
- Always examine the placenta for evidence tubercles as their presence may implicate vertical T

- If the mother is known or suspected of having TB disease, a maternal evaluation to rule out pulmonary or extra pulmonary disease including uterine TB needs to be done
- HIV serology should be undertaken if her prenatal screening is not done or is unavailable
- A BCG vaccine should not be given to a newborn
- An evaluation for congenital TB should be carried out if the newborn is symptomatic or in the cases where the mother is still acid fast bacilli (AFB)-positive or has disseminated disease that was not treated or was partially treated

- An evaluation of a new born include: detailed physical examination, CBC, LFT, ESR, CXR, gastric aspirates for AFB and a TB culture of 3 early morning consecutive samples, an abdominal ultrasound or a lumbar puncture followed by submitting, cerebrospinal fluid for AFB and TB cultures

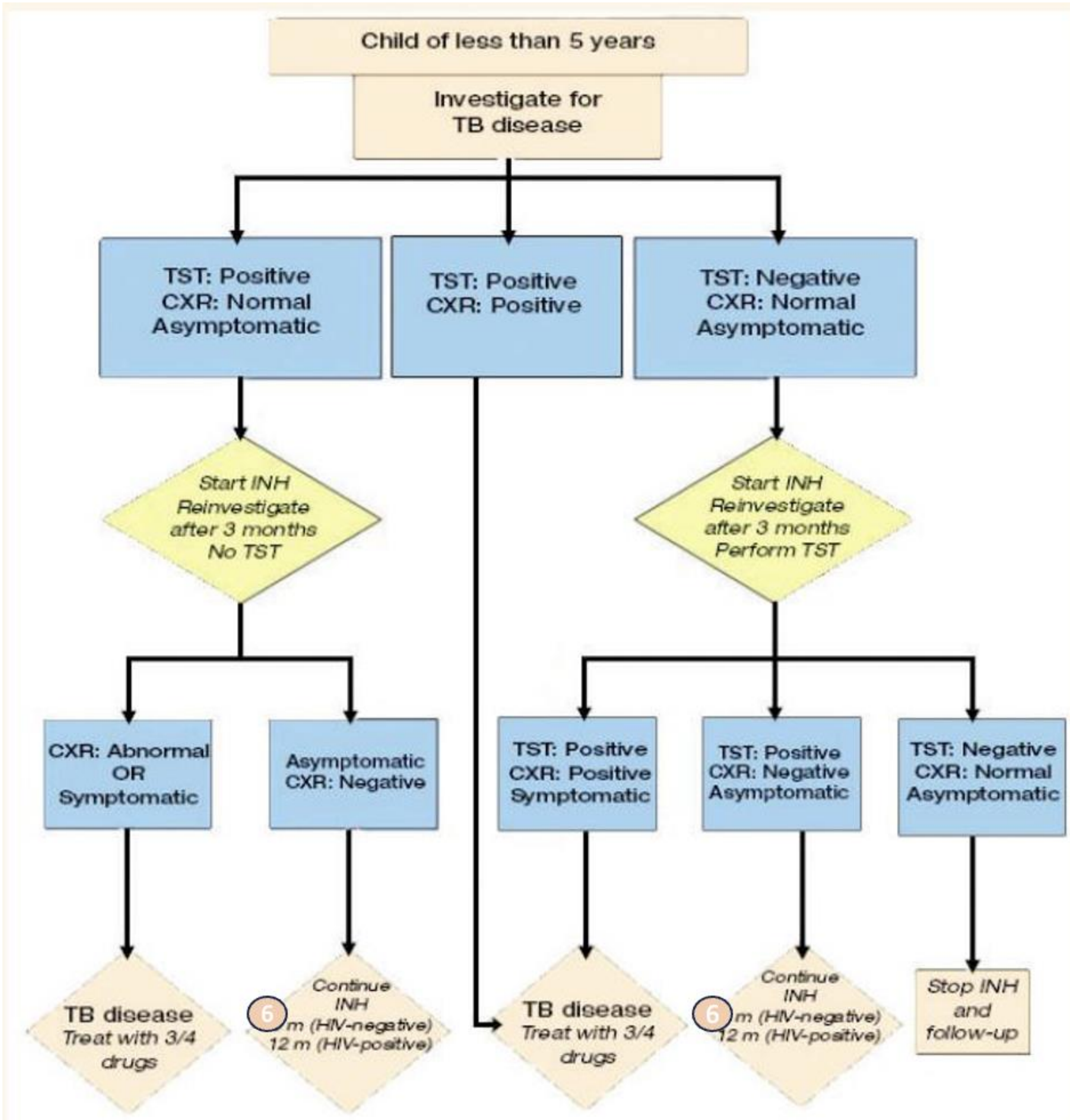
- Contact paediatric infectious diseases specialist.
- If the newborn has evidence of congenital TB, start him/her on H+R+Z+ ethambutol or amikacin for 2 months followed by HR for 6-12 months depend on the disease.
- If congenital TB is excluded start the newborn on INH Continue INH for 3 months on monthly follow up evaluation.
- At 3 months of age, do Mantoux test, CBC, ESR & chest x-ray. If Mantoux test is positive, reassess baby for active TB; IF there is no evidence of tuberculosis disease, treat as LTBI with daily INH for 6 months.
- If Mantoux test is negative & the mother has become AFB negative (non- contagious), INH is discontinued.
- Delay BCG vaccine till 2 weeks after completion of TB prophylaxis or treatment

<u>Separating baby from mother:</u>

Newborn should be separated from any active TB case in the household including his/her mother during the evaluation period and any mother/adult contact should wear a facemask while handling the baby while following the isolation precautions. Once the baby is on INH and the mother or adult source is on full treatment and sputum is AFB negative, there is no need to separate the baby, and the mother can directly breastfeed the baby

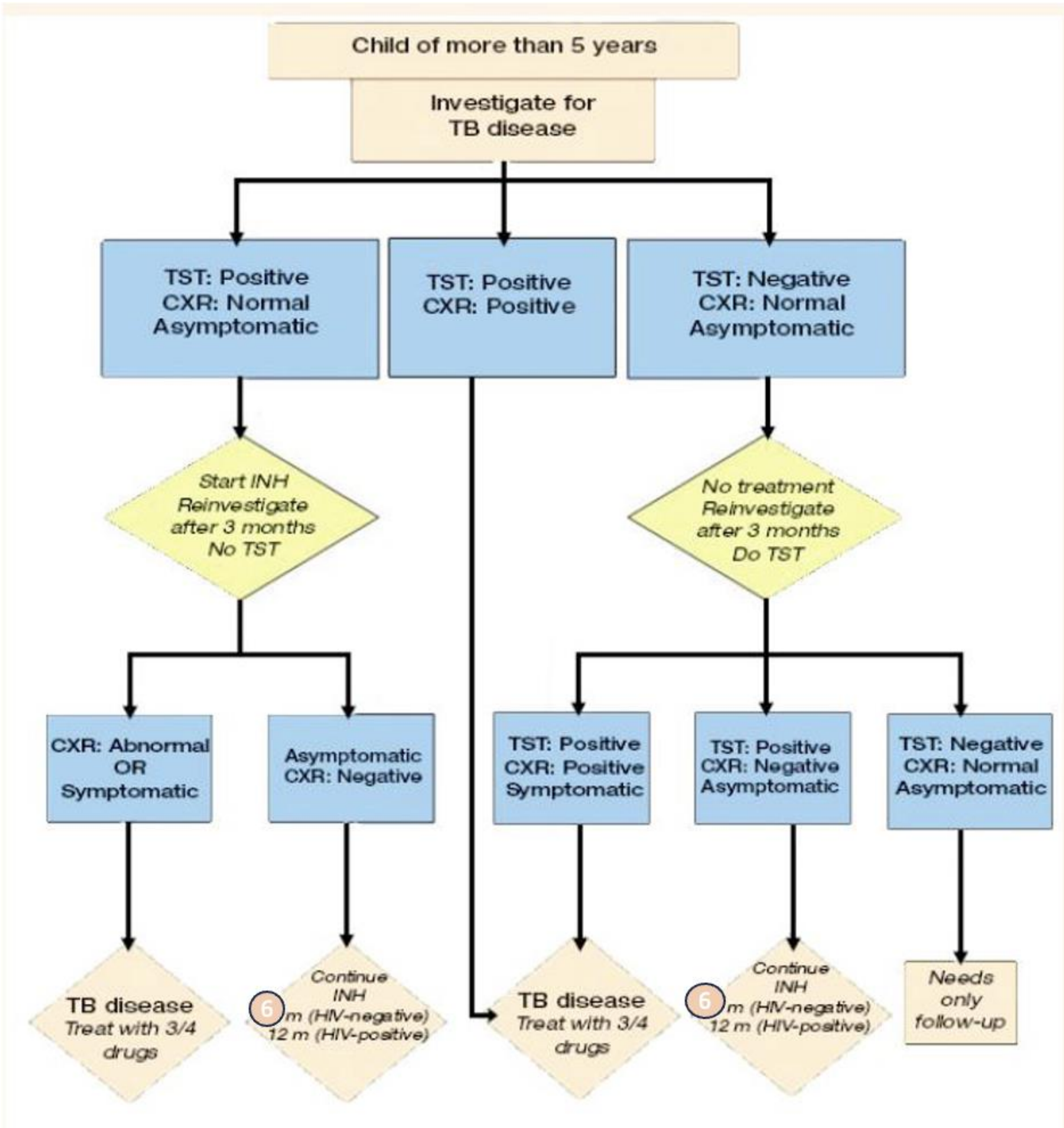
Management of Children under 5 Years:

Follow flowchart below:



Management of Children above 5 Years:

See flowchart below:



- In TB contact children above 5 years of age with immunity impairment, even if the initial TST negative, they should started on INH
- In HIV infected children, duration of latent TB treatment should be of 12 months period

Children and Multidrug Resistant Tuberculosis:

- In a child with exposure to an adult with MDRTB, optimal therapy for those with LTBI is not known.
- Multidrug regimens have been used that can: include pyrazinamide, fluroquinolone, and ethambutol depending on the susceptibility of the isolated organisms.
- Consultation of infectious disease experts for the management of such children is recommended.

Therapies:

- INH is drug commonly used for latent TB treatment with six months regimen
- Standard recommended dose being 10–15 mg/Kg/day (maximum dose: 300 mg/ day).
- INH is well absorbed orally and is best absorbed when consumed on an empty stomach.
- The primary possible adverse reactions include hepatitis (age related), peripheral neuropathy and hypersensitivity reactions
- Other possible adverse effects of INH: optic neuritis, arthralgia, nervous system changes, drug-induced lupus, diarrhea and cramping with liquid product.
- In 10–20% of patients, aminotransferase levels will elevate 2–3 times during the first months of therapy; these levels will usually return to normal despite the continuance of the drug.
- INH should be stopped if liver enzymes 5 times upper limit without symptoms or 3 times upper limit with symptoms

- Vitamin B6(pyridoxine): should be used in situations where high-dose INH is employed, and in the case of children with diabetes, uremia, HIV infection, malnutrition, or peripheral neuropathy, breastfed infants should receive vitamin B6 while taking INH
- Dose of pyridoxine is 1-2 mg/kg/day PO

- **Other Treatment Regimens for TBI:**
- <u>Once-Weekly Dosing of Isoniazid and Rifapentine (3HP) (12 Doses):</u>

  - It is not recommended for Children younger than 2 years of age
  - Patient with HIV/AIDS who are taking antiretroviral medications with clinically significant or unknown drug interactions with once-weekly RPT*
  - People presumed to be infected with INH- or RIF-resistant M. tuberculosis

- <u>4 Months of Daily Rifampin (4R) (120 Doses)</u>
- <u>3 Months of Daily Therapy With Isoniazid and Rifampin (3HR) (90 Doses)</u>

End of therapy assessment:

Families should be informed that immunologic memory will cause either the TST or IGRA test of infection to stay positive after treatment has finished; this is not an indication that the treatment is not working.

What should be done in the clinic?

1- Detailed history of the patient including area of residency
2- Detailed history of index case
3- Date of last contact with index case

4- Detailed physical exam
5- Initial investigations: CBC, ESR, LFT, TST, CXR, QUANTAFERON
6- Reporting form to be filled
7- To repeat TST after 10-12 weeks (3 Months) if initial one is negative and to start children below five years on INH till the next test
8- Detailed parents education regarding possible side effects of medication
9- Two weeks followed up visit to follow on the tolerability of the medication, to look for possible side effects, and to address any parents concerns
10- To ensure that parents should report to a nearest health facility if child develop any of the following: abdominal pain, vomiting, lethargy or jaundice and to stop the medications immediately
11- Make sure that other children and adults in family have been screened to avoid missing other infectious cases in household

Drug	Dose (mg/kg/d ay)	Maximum dose	Adverse reactions
Isoniazid (H)**	10 mg/kg (rang 10–15 mg/kg)	300 mg/day	Mild of AST/ALT, peripheral neuritis, hypersensitivity. The concomitant use of Rifampicin increases incidence of hepatotoxicity.
Rifampicin (R)	15mg/kg (range10–20mg/kg)	600 mg/day	Orange discoloration of urine, contact lenses or secretions, vomiting, hepatitis, influenza like reaction, thrombocytopenia, pruritus, oral contraceptives may be ineffective
Pyrazinamide (Z):	35 mg/kg (range 30–40 mg/kg).	2 g	Hepatotoxicity, GI upset hyperuricemia and arthralgia.
Ethambutol (E)	20 mg/kg (range 15–25 mg/kg).	2.5 g	Optic neuritis (usually reversible), decreased red-green color discrimination, GIT disturbances and hypersensitivity.

Reference:

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
1-Canadian TB manual, 8 t h E d i t i o n	Public Health Authority	2022	Chapter 9
National Tuberculosis Manual Fifth Edition – 2022	Ministry of Health	2022	Chapter 5&6
Tuberculosis Infection in Children and Adolescents: Testing and Treatment	American academy of pediatrics	DOI: https://doi.org/10.1542/peds.2021-054663 PEDIATRICS (ISSN Numbers: Print, 0031-4005; Online, 1098-4275). Copyright © 2021 by the American Academy of Pediatrics	